Reg. No. | | | | | | | | | | |

[12IS753]

SEVENTH SEMESTER B.E EXAMINATION DECEMBER –2015

DISTRIBUTED SYSTEMS

Time:3 Hours **Maximum Marks:70**

Instruction: Answer any **FIVE** full Questions.

1.	**a.**	What is distributed system? Illustrate with suitable examples.	**06**
	b.	Discuss briefly about the challenges of distributed systems.	**08**
2.	**a.**	Describe the characteristics of inter process communication.	**04**
	b.	Define marshalling. Explain the different approaches in marshalling.	**04**
	c.	Write a note on: i. CORBA's common data representations (CDR) ii. Remote object references	**06**
3.	**a.**	i. Explain briefly group communication.	**04**
		ii. Distinguish different RPC communication protocols.	**04**
	b.	Discuss about the IPC in UNIX.	**06**
4.	**a.**	What is cryptography? Explain briefly about the digital signature with a neat diagram.	**07**
	b.	Explain the system architecture of Kerberos.	**07**
5.	**a.**	Explain the file service architecture with neat diagram.	**10**
	b.	Discuss about the design characteristics of Andrew file system.	**04**
6.	**a.**	Explain the central server algorithm to achieve the mutual exclusion.	**07**
	b.	What is Election algorithm? Suppose that two processes detect the demise of the coordinator simultaneously and both decide to hold on election using the bully algorithm. What happens?	**07**
7.	**a.**	Explain the flat and nested distributed transaction with a neat diagram.	**06**
	b.	Discuss about the transaction recovery in detail.	**08**
8.	**a.**	Explain the implementation approaches of distributed shared memory.	**07**
	b.	Explain briefly about the Munin DSM design.	**07**

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